



KILLARA HIGH SCHOOL

YEAR 11 EXAMINATIONS

TERM 4 2005 (HSC)

MATHEMATICS EXTENSION 1

Time Allowed: 70 minutes

Instructions:

- * Attempt ALL questions**
- * Show all necessary working**
- * Marks may NOT be awarded for careless or inadequate working**
- * Board approved calculators may be used**
- * Start each question on a new page**

Question 1 - Application of Series (12 marks)

- (a) Kim was born on 17th August 1975. On this day his father deposited \$500 in an account earning 9% pa interest. On each subsequent birthday Kim's father deposited another \$500. If Kim's father withdrew all the money in the account the day before Kim's 21st birthday and gave it to Kim as a gift, how much did Kim receive?
- (b) A ball is dropped from a height of 8m and bounces to one third of its previous height at each bounce. What is the maximum distance through which the ball would bounce before coming to rest?

Question 2 - Locus and Parabola (12 marks)

- (a) The chord of the parabola $x^2 = 4ay$ joining $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ meets the focal chord parallel to the x -axis at L. Show that the coordinates of L are $(\frac{2a[1+pq]}{p+q}, a)$
- (b) The normals to the parabola $x^2 = 4ay$ at the points $T(2at, at^2)$ and $S(2as, as^2)$ meet at R.
- (i) Show that the coordinates of the point R are $(-ats[t+s], a[t^2+ts+s^2+2])$.
- (ii) If $st = -2$ find the cartesian equation of the locus of R.

Question 3 - Geometric Applications of Calculus (16 marks)

- (a) Given $\frac{d^2 s}{dt^2} = 3$ and that when $\frac{ds}{dt} = 5$, $t = 0$ and $s = 0$, find s in terms of t .
- (b) In a book the pages are to have an area of 338cm^2 . A margin of 1cm is left at each side and 2cm at the top and bottom of each page. If the width of the page is x cm show that the area $A\text{ cm}^2$ of the print is given by $A = 346 - 4x - \frac{676}{x}$. Hence find the dimensions of the page so that there is a maximum area of print.
- (c) For the curve $y = x^3 + 3x^2 - 9x - 8$
- (i) Find the stationary points and distinguish between them.
 - (ii) Find the point of inflexion.
 - (iii) Sketch the curve.

Question 4 - Induction (8 marks)

- (a) Use induction to prove that $3^{3n} + 2^{n+2}$ is divisible by 5, for positive integral values of n .
- (b) Prove by induction, for positive integral values of n that
- $$1 + 3 + 7 + 15 + \dots + (2^n - 1) = 2^{n+1} - n - 2.$$

QUESTION 1

(a) $R = 9\%$ $A_0 = \$500 \dots A_{20} = \500 $n = 21$

@ 21 $A_0 = 500 (1.09)^{21} = T_{21}$

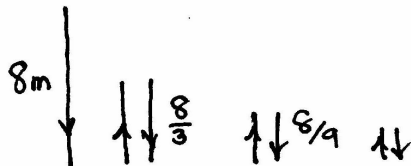
$A_{20} = 500 (1.09) = T_1$

$S_{21} = 500 (1.09) \left[\frac{(1.09)^{21} - 1}{1.09 - 1} \right]$

Kim receives = \$30 936.67

1 mk units
ie \$ and
nearest cent

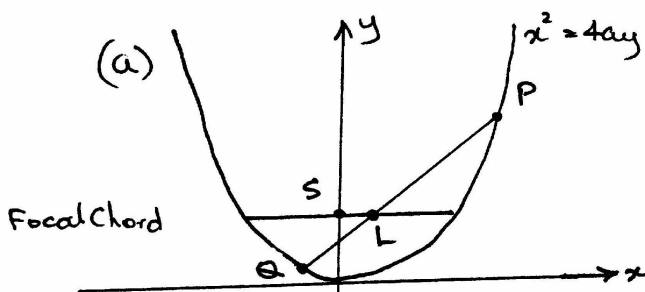
(b)



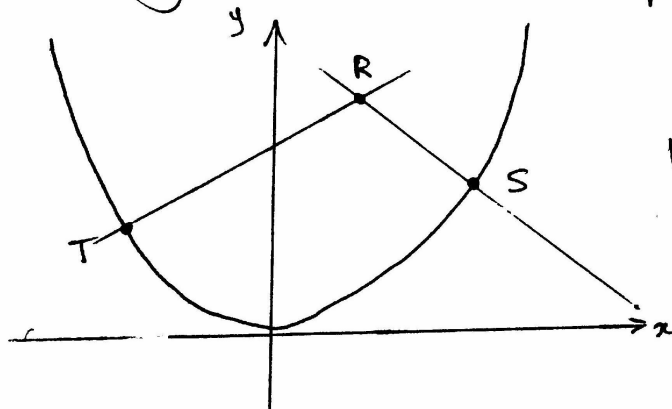
$a = 8$ $r = \frac{1}{3}$ $S_{\infty} = \frac{8}{1 - 1/3} = 12m$

Distance travelled = $2 \times 12 - 8 = 16m$

QUESTION 2 LOCUS and PARABOLA



Focus (0, a) $\therefore$ Chord $y = a$
@ L $y = a$



$P(2ap, ap^2)$ & $Q(2aq, aq^2)$

Chord $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$

$\frac{y - ap^2}{x - 2ap} = \frac{aq^2 - ap^2}{2aq - 2ap}$

$\frac{y - ap^2}{x - 2ap} = \frac{(q - p)(q + p)}{2(q - p)}$

$\therefore y - ap^2 = \frac{1}{2}(q + p)(x - 2ap)$

$y = \left(\frac{p+q}{2}\right)x - apq - p^2 + ap^2$

$y = \left(\frac{p+q}{2}\right)x - apq$

@ $y = a$ $a + apq = \left(\frac{p+q}{2}\right)x$

$2a(1 + pq) = (p + q)x$

$x = \frac{2a(1 + pq)}{p + q}$

$\therefore L$ is $\left(\frac{2a(1 + pq)}{p + q}, a\right)$

$$(b) \quad x^2 = 4ay \quad T(2at, at^2) \quad S = (2as, as^2)$$

Tangent @ T $\left. \begin{aligned} y &= \frac{x^2}{4a} \\ y' &= \frac{2x}{4a} \\ y &= \frac{x}{2a} \\ \therefore -\frac{1}{m} &= -\frac{2a}{x} \end{aligned} \right\} \text{Deriving}$

@ $f'(2at) = -\frac{1}{t}$
 $y - at^2 = -\frac{1}{t}(x - 2at)$
 $xy - at^3 = -x + 2at$

$$x + ty = at^3 + 2at \quad \text{--- (1)}$$

@ S $x + sy = as^3 + 2as \quad \text{--- (2)}$

① - ② $y(t-s) = a(t^3 - s^3) + 2a(t-s)$
 $y = a(t^2 + ts + s^2 + 2)$

Sub in ①

$$x + at(t^2 + ts + s^2 + 2) = at^3 + 2at$$

$$x = -ats(t+s)$$

$$\therefore R \text{ is } (-ats[t+s], a(t^2 + ts + s^2 + 2))$$

2(b)(ii) $x = ats(t+s)$
 $y = a(t^2 + ts + s^2 + 2)$

If $ts = -2 \quad \therefore \left. \begin{aligned} x &= -2a(t+s) \\ (t+s) &= -\frac{x}{2a} \end{aligned} \right\} 1$

$\therefore y = a[(t+s)^2 - ts + 2]$
 $y = a\left[\left(-\frac{x}{2a}\right)^2 + 2 + 2\right] \quad 1$

$$y = \frac{x^2}{4a^2} + 4a \quad 1$$

$$y = \frac{x^2}{4a} + 4a$$

or $4ay = x^2 + 16a^2$

or $x^2 = 4a(y - 4a) \quad \left. \right\} 1$

QUESTION 3.

(a) $\frac{d^2s}{dt^2} = 3$

$$\therefore \frac{ds}{dt} = 3t + k$$

But $s = 0 + k$

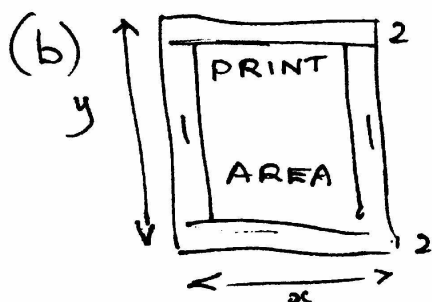
$$\therefore k = 5$$

So $ds/dt = 3t + 5$

$$s = \frac{3}{2}t^2 + 5t + C$$

$$0 = 0 + 0 + C$$

$$\therefore s = \frac{3}{2}t^2 + 5t$$



$$A = 338 = x \times y$$

$$\therefore y = 338/x$$

$$\begin{aligned} \text{Print Area} &= (x-2) \times (y-4) \\ &= (x-2) \left(\frac{338}{x} - 4 \right) \\ &= 338 - 4x - \frac{676}{x} + 8 \end{aligned}$$

$$A = 346 - 4x - \frac{676}{x}$$

$$\therefore A' = -4 + \frac{676}{x^2}$$

Let $A' = 0$

$$\therefore 4x^2 = 676$$

$$x^2 = 169$$

$$x = 13$$

$$y = \frac{338}{13}$$

$$= 26$$

Dimensions $13\text{cm} \times 26\text{cm}$.

QUESTION 3.

7 (c) $y = x^3 + 3x^2 - 9x - 8$

(i) $y' = 3x^2 + 6x - 9$

Let $y' = 0$. $\therefore x^2 + 2x - 3 = 0$

$(x+3)(x-1) = 0$

$x = -3$ or $x = 1$

$y'' = 6x + 6$

Let $y'' = 0$ $\therefore x = -1$

$y''(-3) = -18 + 6 < 0 \therefore \text{max}$

$y''(1) = 6 + 6 > 0 \therefore \text{min.}$

$f(-3) = -27 + 27 + 27 - 8 = 19$

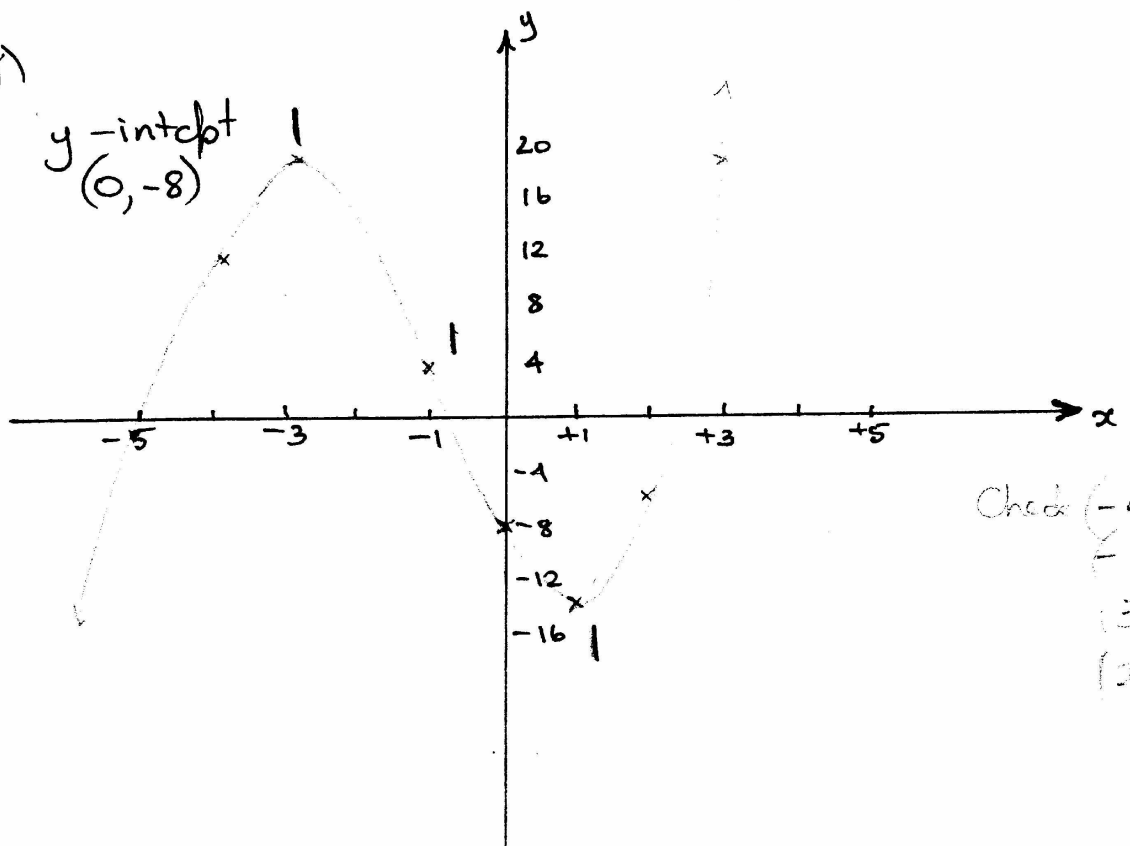
$f(1) = 1 + 3 - 9 - 8 = -13$

$f(-1) = -1 + 3 + 10 - 8 = 4$

$\therefore \text{Min @ } (1, -13) \text{ Max @ } (-3, 19)$

(ii) Point of inflexion $(-1, 4)$.

(iii)



Check $(-4, 12)$
 $(-5, -1)$
 $(2, 10)$
 $(3, -6)$

QUESTION 4

(a) $3^{3n} + 2^{n+2}$ divisible by 5

| Step 1 For $n=1$ LHS = $3^3 + 2^3$
 $= 27 + 8$
 $= 35$
 $= 5 \times 7$

$\therefore$ True for $n=1$.

| Step 2 Assume $n=k$.
 i.e. $3^{3k} + 2^{k+2} = 5 \times m$. (m an integer)

| Step 3 Prove for $n=k+1$
 i.e. $3^{3(k+1)} + 2^{(k+1)+2} = 5p$. (p an integer)

$$\begin{aligned} \text{LHS} &= 3^{3k+3} + 2^{k+3} \\ &= (5m - 2^{k+2}) \times 27 + 2^{k+3} \\ &= 5 \times 27m - 2^k \times 4 \times 27 + 2^{k+3} \\ &= 5 \times 27m - 2^k (108) + 2^{k+3} \\ &= 5 \times (27m - 20 \times 2^k) \\ &\therefore \div 5 \text{ or } = 5p. \end{aligned}$$

| Step 4 $\therefore$ If true for $n=k$ then true for $n=k+1$
 But true for $n=1 \therefore$ true for $n=2$
 Now true for $n=2 \therefore$ " " $n=3$
 and so on for all integer $n \geq 1$.

(b) $1 + 3 + 7 + 15 + \dots + (2^n - 1) = 2^{n+1} - n - 2$.

| Step 1 For $n=1$ LHS = 1 RHS = $2^2 - 1 - 2 = 1 = \text{LHS}$
 $\therefore$ True for $n=1$.

| Step 2 Assume for $n=k$
 i.e. $1 + 3 + 7 + 15 + \dots + (2^k - 1) = 2^{k+1} - k - 2$

| Step 3 Prove for $n=k+1$
 i.e. $1 + 3 + 7 + 15 + \dots + (2^{k+1} - 1) = 2^{k+2} - (k+1) - 2$

$$\begin{aligned} \text{LHS} &= (1 + 3 + 7 + \dots + 2^k - 1) + (2^{k+1} - 1) \\ &= 2^{k+1} - k - 2 + 2^{k+1} - 1 \\ &= 2 \times 2^{k+1} - k - 3 \\ &= \text{RHS} \end{aligned}$$

| Step 4 $\therefore$ If true for $n=k$ then true for $n=k+1$
 But true for $n=1 \therefore$ true for $n=2$
 and so on for all integers n .